h-adic topology and profinite linear lambda terms

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1 Profinite Lambda terms and related concepts

One starts with the observation that if we have a work over a finite alphabet Σ:

$$W = a_{w_1} \dots a_{w_n} \in \Sigma^*$$

then one can represent W as a well-typed term in the simply typed lambda calculus:

$$a_{w_1} : q \to q, \dots, a_{w_n} : q \to q \vdash W : q \to q$$

where q is a base type that we informally interpret as a type of states, so we have a closed lambda term

$$\lambda a_{w_1}^{q \to q} \dots \lambda a_{w_n}^{q \to q} \lambda x^q . a_{w_k}(\dots(a_{w_1}\ q)\dots) : (q \to q) \dots (q \to q) \to q \to q$$

encoding a word w via the instance of the Church encoding. Further, we will call the type of such terms Church_Σ. Further, we have the monoid isomorphism:

$$\Sigma^* \cong \{M \text{ is a lambda term } | M : \text{Church}_\Sigma \}/\beta\eta$$

In other words, words over a finite alphabet can be viewed as lambda terms of a particular type factorised module the $\beta\eta$-equivalence. Generally, there is amount literature studying both syntactic and semantic aspects of the Church encoding of the finite words.

The claim made in [vGMM23] connects the finitary interpretation of STLC in the category of finite sets **FinSet** with deterministic finite automata as follows. Take any finite set Q, so for any type A, we define a set $[\![A]\!]_Q$ inductively defined as

- $[\![p]\!]_Q = Q$
- $[\![A \to B]\!]_Q = [\![A]\!]_Q \Rightarrow [\![B]\!]_Q$

so every term $x_1 : A_1, \ldots, x_n : A_n \vdash M : A$ transports to the function between finite sets $[\![M]\!] : [\![A_1]\!]_Q \times \ldots \times [\![A_n]\!]_Q \to [\![A]\!]_Q$. On closed terms, the interpretation induces a function $[\![.]\!]_Q : \Lambda_{\beta\eta}(A) \to [\![A]\!]_Q$ mapping every closed term M of type A to its interpretation $[\![M]\!]_Q$. Further, a word $W = a_{w_1} \ldots a_{w_k} \in \Sigma^*$ is mapped to the following morphism in **FinSet**: $[\![W]\!]_Q : (Q \Rightarrow Q) \times \ldots \times (Q \Rightarrow Q) \to (Q \Rightarrow Q)$, i.e.,

$$[\![W]\!]_Q : f_{w_1}, \ldots, f_{w_k} \mapsto f_{w_k} \circ \ldots \circ f_{w_1}$$

Recall that a *deterministic finite state automaton* on a finite alphabet $\Sigma = \{a_1, \ldots, a_n\}$ is a structure $\mathcal{A} = (Q, \delta, q_0, Acc)$, where Q is a finite set of states, $\delta : \Sigma \times Q \to Q$ is a transition function, $q_0 \in Q$ is the initial state and $Acc \subseteq Q$ is the set of accepting states. The transition function δ induces a family of functions $\delta_{a_i}(q_k) = \delta(a_i, q_k) : Q \to Q$ for each $a_i \in \Sigma$ and $q_k \in Q$. $\delta_{a_i}(q_k) = q_{k'}$ means that $\mathcal{A}$ transitions from q_k to q'_k encountering a_i. Now one can apply the interpretation $[\![W]\!]_Q$ to the functions $\delta_{a_{w_i}}$ for $1 \leq i \leq k$, so we obtain

$$\delta_w : [\![W]\!]_Q(\delta_{a_{w_1}}, \ldots, \delta_{a_{w_k}}) : Q \to Q$$

Therefore, given a deterministic finite automaton $\mathcal{A} = (Q, \delta, q_0, Acc)$, the tuple (Q, δ, q_0) induces the evaluation function

$$\mathrm{eval}_{\delta,q_0} : [\![\mathrm{Church}_\Sigma]\!]_Q \to Q$$

transporting every functional $F \in [\![\mathrm{Church}_\Sigma]\!]_Q$ to the state $F(\delta_{a_{w_1}}, \ldots, \delta_{a_{w_k}})$ in Q, therefore we have

$$\Sigma^* \cong \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma) \to [\![\mathrm{Church}_\Sigma]\!]_Q \xrightarrow{\mathrm{eval}_{\delta,q_0}} Q$$

so we associate every finite word $w \in \Sigma^*$ with the final state $q_f = \delta_w(q_0)$. The inverse image $Acc \subseteq Q$ under this composite is called a *regular language* $L_\mathcal{A}$, the set of finite words recognisable by $\mathcal{A}$.

Regular languages $L_\mathcal{A}$ form a Boolean algebra $\mathrm{Reg}_Q(\mathrm{Church}_\Sigma)$ of regular languages of lambda terms of type Church_Σ recognised by Q. The algebra can be defined as the image of the Boolean algebra homomorphism $[\![.]\!]_Q^{-1} : 2^{[\![\mathrm{Church}_\Sigma]\!]} \to 2^{\Sigma^*}$ obtained by applying the contravariant powerset functor $2^{(.)} : \mathbf{FinSet}^{op} \to \mathbf{BA}$ to the semantic bracket $[\![.]\!]_Q$. The Boolean algebra $\mathrm{Reg}_Q(A)$ of regular languages of lambda terms of any higher order type A recognisable by a finite set Q is defined as the image of the Boolean algebra homomorphism $[\![.]\!]_Q^{-1} : 2^{[\![A]\!]_Q} \to 2^{\Lambda_{\beta\eta}(A)}$. That is, a set L of lambda terms of type A is recognisable by the finite set Q exactly when it has the form $[\![.]\!]_Q^{-1}(\mathrm{Acc}) = \{M \in \Lambda_{\beta\eta}(A) \,|\, [\![M]\!]_Q \in \mathrm{Acc}\}$ for some choice $\mathrm{Acc} \subseteq [\![A]\!]_Q$ of accepting elements. Thus, the collection $\mathrm{Reg}(A) \subseteq 2^{\Lambda_{\beta\eta}(A)}$ of lambda terms of type A as follows

$$\mathrm{Reg}(A) = \bigcup \{\mathrm{Reg}_Q(A) \,|\, Q \in \mathbf{FinSet}\}.$$

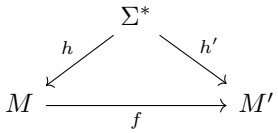
The monoid $\widehat{\Sigma^*}$ is the free profinite monoid generated by Σ. $\widehat{\Sigma^*}$ can be constructed as the limit, computed in the category of monoids **Mon**, of the projective system of finite monoid homomorphisms

$$(\Sigma^*/\phi \to \Sigma^*/\phi')_{\phi \subseteq \phi'}$$

where ϕ and ϕ' are finite index congruences on the free monoid Σ^* such that $\phi \subseteq \phi'$. Every finite index congruence ϕ, in turn, induces a surjective homomorphism $h : \Sigma^* \to M$ for the quotient monoid $M = \Sigma^*/\phi$. The surjectivity condition on h can be relaxed to show that $\widehat{\Sigma^*}$ of profinite words is the codirected limit of the composite functor

$$\Sigma^*/\mathbf{FinMon} \xrightarrow{\pi} \mathbf{FinMon} \hookrightarrow \mathbf{Mon}$$

$\Sigma^*/\mathbf{FinMon}$ is the slice category whose objects (M, h) are pairs consisting of a finite monoid M and a homomorphism $h : \Sigma^* \to M$ and whose morphisms $(M, h) \to (M', h')$ are homomorphisms $f : M \to M'$ such that the following triangle commutes:



The projection functor $\pi : \Sigma^*/\mathbf{FinMon} \to \mathbf{FinMon}$ transports (M, h) to its underlying monoid M. One obtains $\widehat{\Sigma^*}$ as the limit of a codirected diagram of monoid homomorphisms $(M \to M')_{(M,h)\to(M',h')}$ extending the diagram formed from the family $(\Sigma^*/\phi \to \Sigma^*/\phi')_{\phi\subseteq\phi'}$ to all homomorphisms $h : \Sigma^* \to M$ to a finite monoid M.

Recall that every homomorphism $h : \Sigma^* \to M$ induces a deterministic finite automaton with $Q := M$, $\delta(a, q) = q \cdot_M a$ for every $q \in M$ and $a \in \Sigma$. So every homomorphism $h : \Sigma^* \to M$ (assume $M = \{q_1, \ldots, q_m\}$) induces a decomposition of Σ^* into m components $L_{q_i} = h^{-1}(q_i)$ for $1 \leq i \leq m$ where each L_{q_i} is a regular language. Let $\mathrm{Reg}_{(M,h)}(\Sigma)$ denote the Boolean algebra of languages of the form $L_q = h^{-1}(q)$ for $q \in M$. We have a functor:

$$\mathrm{Reg}_{(.)}(\Sigma) : (\Sigma^*/\mathbf{FinMon})^{op} \to \mathbf{BA}$$

The Boolean algebra $\mathrm{Reg}_{(M,h)}(\Sigma)$ coincides with 2^h, the image of $h : \Sigma^* \to M$ applied to the contravariant powerset functor. Also, the following directed diagram in $\mathbf{BA}$ associated to $\mathrm{Reg}_{(.)}(\Sigma)$:

$$(\mathrm{Reg}_{(M',h')}(\Sigma) \to \mathrm{Reg}_{(M,h)}(\Sigma))_{(M,h)\to(M',h')}$$

can be obtained by applying the contravariant powerset functor to the above trianlge and $(M \to M')_{(M,h)\to(M',h')}$.

One moves from profinite words to profinite lambda terms in the following way. For any finite set Q, one has the interpretation of any type A as the finite set $[\![A]\!]_Q$. Given another finite set Q' with a relation $R \subseteq Q \times Q'$, one can relate different interpretations of A by constructing a relation $[\![A]\!]_R \subseteq [\![A]\!]_Q \times [\![A]\!]_{Q'}$. Such relations are called *logical relations*. Lambda terms are parametric: that is, if M has type A and $R \subseteq Q \times Q'$, then $([\![M]\!]_Q, [\![M]\!]_{Q'}) \in [\![A]\!]_R$.

Now let us discuss how one can define the exponential of relations. Let $S \subseteq P \times P'$ and $R \subseteq Q \times Q'$ be relations on finite sets, the exponential $S \Rightarrow R \subseteq (P \Rightarrow Q) \times (P' \Rightarrow Q')$ is given as:

$$S \Rightarrow R := \{(g, h) \mid \forall x \in P \forall y \in P' \; (xSy \to g(x)Rh(y))\}$$

Therefore, for any relation $R \subseteq Q \times Q'$ between finite sets Q and Q', one constructs a relation $[\![A]\!]_R \subseteq [\![A]\!]_Q \times [\![A]\!]_{Q'}$ defined by induction on A as follows:

- $[\![p]\!] = R$,
- $[\![A \Rightarrow B]\!]_R = [\![A]\!]_R \Rightarrow [\![B]\!]_R$

The *fundamental lemma of logical relations* states that for any $M \in \Lambda_{\beta\eta}(A)$ for any A and for any $R \subseteq Q \times Q'$, the interpretations of M are related as follows:

$$([\![M]\!]_Q, [\![M]\!]_{Q'}) \in [\![A]\!]_R$$

Further, we will be using partial surjections since they are stable under taking relational exponentials. Also, if $f : Q \twoheadrightarrow Q'$ is a partial surjection, so $[\![A]\!]_f \subseteq [\![A]\!]_Q \times [\![A]\!]_{Q'}$ for any A. Besides, we have the inclusion of Boolean algebras $\mathrm{Reg}_{Q'}(A) \subseteq \mathrm{Reg}_Q(A)$, which is proved using the fundamental lemma of logical relations.

Theorem 1.1. The diagram of Boolean algebras $\mathrm{Reg}_{(.)}(A) : \mathbf{FinPSurj}^{op} \to \mathbf{BA}$:

$$(\mathrm{Reg}_{Q'}(A) \hookrightarrow \mathrm{Reg}_{Q}(A))_{f:Q \to Q' \in \mathbf{FinPSurj}}$$

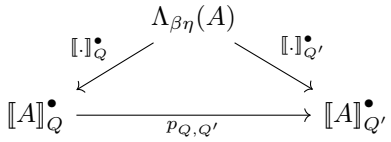
is directed, and its colimit in **BA** coincides with the Boolean algebra $\mathrm{Reg}(A)$ of regular languages for a higher-order type A.

Also, for any simple type A, the set of regular languages $\mathrm{Reg}(A)$ is closed under finite intersections.

Now let us discuss how the space of profinite lambda terms is constructed. The idea is that for any simple type A as parametric families of semantic elements, so they form a Stone space dual to the Boolean algebra $\mathrm{Reg}(A)$.

Proposition 1.1. Let Q, Q' be finite sets,

1. The set of atoms $\mathrm{At}_Q(A)$ of the finite Boolean algebra $\mathrm{Reg}_Q(A)$ is in bijection with the set $[\![A]\!]^\bullet_Q$ of definable elements of simple type A, given by the function $q \mapsto [\![.]\!]_Q^{-1}(\{q\}) : \mathrm{At}_Q(A) \to [\![A]\!]^\bullet_Q$.
2. For partial surjection $f : Q \twoheadrightarrow Q'$ and for any simple type A, the set $[\![A]\!]^\bullet_Q \subseteq [\![A]\!]_Q$ of definable elements is contained in the domain of $[\![A]\!]_f$, and the restriction of $[\![A]\!]_f$ to $[\![A]\!]^\bullet_Q$ is the unique function $p_{Q,Q'}$ such that the following triangle commutes:



Note that if $f, g : Q \to Q'$ are partial surjections, then, while the semantic interpretations $[\![A]\!]_f, [\![A]\!]_g : [\![A]\!]_Q \to [\![A]\!]'_Q$ are different, their restrictions to $[\![A]\!]^\bullet_Q$ are both equal to $p_{Q,Q'}$.

One defines profinite lambda terms of simple type A as follows.

Definition 1.1. Let A be a simple type, the set of *profinite lambda terms* is defined as the limit $\widehat{\Lambda}_{\beta\eta}(A)$ in **Set** of the diagram:

$$([\![A]\!]^\bullet_f : [\![A]\!]^\bullet_Q \twoheadrightarrow [\![A]\!]^\bullet_{Q'})_{f:Q \twoheadrightarrow Q' \in \mathbf{FinPSurj}}$$

2 h-adic topology

The following summary is based on [Kas12, Chapter XVI].

Consider the complex algebra $K = \mathbb{C}[[h]]$ of complex formal series in one variable h. Every element $f \in \mathcal{K}$ has the form

$$f = \sum_{n=0}^{\infty} a_n h^n$$

where $a_n \in \mathbb{C}$ for each $n \in \mathbb{N}$. Every polynomial $f \in \mathbb{C}[h]$ is clearly a formal series. K form a commutative ring with operations given as

- $f + g = \sum\limits_{n=0}^{\infty} a_n h^n + \sum\limits_{n=0}^{\infty} b_n h^n = \sum\limits_{n=0}^{\infty} (a_n + b_n) h^n$

- $f \cdot g = (\sum_{n=0}^{\infty} a_n h^n) \cdot (\sum_{n=0}^{\infty} b_n h^n) = \sum_{n=0}^{\infty} (\sum_{n=p+q} a_p b_q) h^n$

A formal series f is invertible iff its every coefficient is non-zero.

For any $n > 0$, consider the algebra $K_n = \mathbb{C}[h]/(h^n)$ of truncated polynomials obtained as the quotient of the ring of complex polynomials generated by the ideal (h^n). There is a morphism

$$\pi_n : K \to K_n$$

sending each formal series $\sum_{n=0}^{\infty} a_n h^n$ to the class of $\sum_{k=0}^{n-1} a_k h^k$ modulo (h^n). π_n is onto and its kernel is the ideal $h^n K$ generated by h^n in the ring of formal series. Therefore, we have the following isomorphism:

$$\mathbb{C}[[h]]/(h^n) \cong \mathbb{C}[h]/(h^n)$$

For $n > 0$, there is a surjective map of algebras $p_n : K_n \to K_{n-1}$ generated by the inclusion of ideals $(h^n) \subset (h^{n-1})$. So we have the inverse system of algebras $(K_n, p_n)_{n>0}$ and its inverse limit $\varprojlim_n K_n$. One has $p_n \circ \pi_n = \pi_{n-1}$ for each n. Also, there is a unique morphism of algebras from K to $\varprojlim_n K_n$ whose composition with the projection of the inverse limit is equal to π_n. Besides, the map $\pi : \mathbb{C}[[h]] \twoheadrightarrow \varprojlim_n \mathbb{C}[[h]]/(h^n)$ is the isomorphism, which is partly the instance of Birkhoff's subdirect representation theorem, [SB81, Theorem 8.6].

Let us equip K with the *h-adic topology*. The family of of subsets $\pi_n^{-1}(U_n)$ for $n > 0$ and $U_n \subseteq K_n$ form an open basis on K. $\{0\}$ is a family of open neighbourhoods in the discrete set K_n, the family $\pi_n^{-1}(0)$ is a family of open neighbourhoods of 0 in K for the h-adic topology. There is also the structure of a metric space on K given as follows. Given a formal series $f = \sum_{n=0}^{\infty} a_n h^n$, let $o(f)$ denote the unique natural number such that $a_{o(f)} \neq 0$ and $a_k = 0$ for each $k < o(f)$. For $f = 0$, set $o(f) = +\infty$ and extend $(\mathbb{N}, <)$ by adding the top element $+\infty$. It is easy to see that:

$$(h^n) = \{f \in K \mid o(f) > n - 1\}$$

and therefore

$$\bigcap_{n>0} (h^n) = \{0\}$$

We also have for any series $f, g \in K$:

$$o(f + g) \geq \min(o(f), o(g))$$

Now define $|.| : K \to \mathbb{R}_{\geq 0}$ as

$$|f| = 2^{-o(f)}$$

Then if we put $d(f, g) = |f - g|$, then $(\mathbb{C}[[h]], d)$ becomes an ultrametric space.

Let M be a left K-module, consider the family $(h^n M)_{n>0}$ of submodules of M with the canonical K-linear projections:

$$p_n : M_n = M/h^n M \to M_{n-1} = M/h^{n-1} M$$

They form an inverse system of K-modules, so we can consider the inverse limit

$$\widetilde{M} = \varprojlim_n M_n$$

having a natural structure of a left K-module. $\widetilde{M}$ has a natural topology as well, the inverse limit topology, for which the family of submodules $(h^n \widetilde{M})_n$ form an open basis. The projections

$i_n : M \to M_n$ induce a unique K-linear map $i : M \to \widetilde{M}$ such that $\pi_n \circ i = i_n$ for each n. The kernel of i is given as

$$\mathrm{Ker}(i) = \bigcap_{n>0} h^n M$$

A K-module M is *separated* if $\mathrm{Ker}(i) = \{0\}$. It is *complete* if i is injective.

For any module K, the module $M/\mathrm{Ker}(i)$ is separated and the completion $\widetilde{M}$ of M is complete.

Generally, any separated complete K-module M is equipped with the h-adic topology induced by the inverse limit topology on $\widetilde{M}$ via $M \cong \widetilde{M}$.

Now let us describe the following class of separated complete K-modules. Let V be a vector space over $\mathbb{C}$, let $V[[h]]$ denote the set of all formal series in variable h

$$f = \sum_{n=0}^{\infty} a_n h^n$$

where a's range over elements of V. There is a left K-module on $V[[h]]$, and such modules are called *topologically free modules*. One can observe that any topologically free module is separated and complete. Also, if $\{e_i\}_{i\in I}$ is a basis of V, then the K-submodule generated by $\{e_i\}_{i\in I}$ is dense in V for the h-adic topology. Besides, let N be a complete separated K-module, there is a natural isomorphism in the category of K-modules and K-linear maps:

$$\mathrm{Hom}_K(V[[h]], N) \cong \mathrm{Hom}_K(V, N)$$

Further, a K-module M is topologically free iff it is complete, separated and torsion-free, that is, there is such that $mh \neq 0$ for any non-zero $m \in M$.

3 On Dialogue Categories and their internal language

Definition 3.1. A dialogue category is a monoidal category $(\mathcal{C}, \otimes, e)$ equipped with an object $\perp$ along with a family of isomorphisms

$$\varphi_{A,B} : \mathcal{C}(A \otimes B, \perp) \cong \mathcal{C}(B, A \multimap \perp)$$

natural in x and y; along with a family of isomorphisms

$$\psi_{A,B} : \mathcal{C}(A \otimes B, \perp) \cong \mathcal{C}(A, \perp \circ\!\!- B)$$

natural in x and y as well.

3.1 Term Assignment for Tensorial Logic

Figure 1: Term Calculus for Tensorial Logic

$$\frac{}{\vdash \mathbf{1} : \mathbb{1}}\ \mathbb{1}\mathbf{I} \qquad \frac{}{x : A \vdash x : A}\ \mathbf{ax} \qquad \frac{\Gamma \vdash M : \mathbb{1} \qquad \Delta \vdash N : B}{\Gamma, \Delta \vdash \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ N : B}\ \mathbb{1}\mathbf{E}$$

$$\frac{x : A, \Gamma \vdash M : \bot}{\Gamma \vdash \nabla_l x.M : A \multimap \bot}\ \multimap \bot_{\mathbf{I}} \qquad \frac{\Delta \vdash N : A \qquad \Gamma \vdash M : A \multimap \bot}{\Gamma, \Delta \vdash N \triangleleft M : \bot}\ \multimap \mathbf{E}$$

$$\frac{\Gamma, x : A \vdash M : \bot}{\Gamma \vdash \nabla_r x.M : \bot \circ\!\!- A}\ \bot \circ\!\!-_{\mathbf{I}} \qquad \frac{\Gamma \vdash M : A \circ\!\!- \bot \qquad \Delta \vdash N : A}{\Gamma, \Delta \vdash M \triangleright N : \bot}\ \multimap \mathbf{E}$$

$$\frac{\Gamma \vdash M : A \qquad \Delta \vdash N : B}{\Gamma, \Delta \vdash M \otimes N : A \otimes B}\ \otimes\mathbf{I} \qquad \frac{\Gamma \vdash M : A \otimes B \qquad \Delta, x : A, y : B \vdash N : C}{\Delta, \Gamma \vdash \mathbf{let}\ x \otimes y\ =\ M\ \mathbf{in}\ N : C}\ \otimes\mathbf{E}$$

Definition 3.2. Reduction rules for tensorial logic:

$(\multimap \bot\beta)$ $N \triangleleft (\nabla_l x.M) \to M[x := N]$,

$(\bot \circ\!\!- \beta)$ $(\nabla_r x.M) \triangleright N \to M[x := N]$,

$(\mathbb{1}\beta)$ $\mathbf{let}\ \mathbf{1} = \mathbf{1}\ \mathbf{in}\ N \to N$,

$(\otimes\beta)$ $\mathbf{let}\ x \otimes y = M \otimes N\ \mathbf{in}\ P \to P[x := M, y := N]$,

$(\multimap \bot\eta)$ $(\nabla_l x.x \triangleleft M) \to M$ if x is not free in M,

$(\bot \circ\!\!- \eta)$ $(\nabla_r x.M \triangleright x) \to M$ if x is not free in M,

$(\otimes\eta)$ $\mathbf{let}\ x \otimes y = M\ \mathbf{in}\ N[z := x \otimes y] \to N[z := M]$,

$(\otimes_{\mathbf{nat}})$ $P[w := \mathbf{let}\ x \otimes y = M\ \mathbf{in}\ N] \to \mathbf{let}\ x \otimes y = M\ \mathbf{in}\ P[w := N]$,

$(\mathbb{1}_{\mathbf{nat}})$ $P[w := \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ N] \to \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ P[w := N]$.

3.2 Parametricity for Tensorial Logic

The following subsection develops parametricity for the tensorial calculus in the fashion of [AMP25] and [ZZZ10].

References

[AMP25] CB Aberlé, Chris Martens, and Frank Pfenning. Substructural parametricity. *arXiv preprint arXiv:2503.03153*, 2025.

[Kas12] Christian Kassel. *Quantum groups*. Springer Science & Business Media, 2012.

[SB81] Hanamantagouda P Sankappanavar and Stanley Burris. A course in universal algebra. *Graduate Texts Math*, 78:56, 1981.

[vGMM23] Sam van Gool, Paul-André Melliès, and Vincent Moreau. Profinite lambda-terms and parametricity. *Electronic Notes in Theoretical Informatics and Computer Science*, 3, 2023.

[ZZZ10] Jianzhou Zhao, Qi Zhang, and Steve Zdancewic. Relational parametricity for a polymorphic linear lambda calculus. In *Asian Symposium on Programming Languages and Systems*, pages 344–359. Springer, 2010.